

Embedded Systems: India Takes Baby Steps Towards End-to-End Development

India is developing its own embedded design market, moving away from being a strictly outsourced industry. This means that Indian designers and engineers will have to drive actual product innovation and move towards product ownership, which to some extent is already happening



UMA GUPTA

Embodied system development in India started during the early 2000s. According to a report released by National Association of Software and Services Companies (Nasscom) along with Zinnov Management Consulting, the total addressable embedded system R&D global opportunity is expected to reach \$89 billion by 2015. India now has an established base for R&D and can leapfrog both for international and domestic consumption.

The embedded systems market in India has the potential to triple in

size to about \$14.4 billion in 2015 from \$4.95 billion in 2009. Currently, India's revenue from embedded systems development is almost \$5 billion, employing about 110,000 professionals.

End-to-end product development finally happening

Indian companies have migrated from being centres of resource augmentation to product innovation and design centres. Last few years have seen a huge growth in the number of small and innovative companies doing complete design of innovative embedded products. A similar trend exists in the

captive design centres of multinationals and Indian service companies. They are becoming accountable for more of complete product design, delivery and ownership.

"We don't have to look long back to see India as just 'implementer' of finalised design. Till the year 2009, we were just realising systems decided in North America, Europe, etc, which is not the case now. Over a period of time, India has become strategic partner of global product companies aiming to design innovative products within a short span of time. Slowly India has moved up in the value chain and become a design partner that associates right from feasibility study to actual design and launch. We have started managing the complete life-cycle of embedded design," informs Tejas J. Mehta, marketing manager, elfnchips.

Today, almost all MNCs have a significant presence in India, developing embedded design solutions for their worldwide businesses. In addition to having a captive design centre in India, multinationals leverage on the talent pool in multiple third-party houses across India, which provide valuable support in executing part of the design chain activity or developing complete IC design.

Currently, India mainly provides firmware, board design and VLSI design services. There are immense opportunities in hardware and embed-